

■ 1.5.10 Numerical Mathematics

Exact symbolic results are usually very desirable when they can be found. In many calculations, however, it is not possible to get symbolic results. In such cases, you must resort to numerical methods.

<code>N[expr]</code>	numerical value of an expression (see Section 1.1)
<code>NIntegrate[f, {x, xmin, xmax}]</code>	numerical approximation to $\int_{xmin}^{xmax} f \, dx$
<code>NSum[f, {i, imin, Infinity}]</code>	numerical approximation to $\sum_{imin}^{\infty} f$
<code>NRoots[poly==0, x]</code>	numerical approximations to the roots of a polynomial
<code>FindRoot[lhs==rhs, {x, x0}]</code>	search for a numerical solution to the equation, starting with $x=x_0$
<code>FindMinimum[f, {x, x0}]</code>	search for a minimum of f , starting with $x=x_0$

Some numerical operations.

Mathematica cannot get an exact formula for this integral.

```
In[1]:= Integrate[Sin[Sin[x]], {x, 1, 2}]
Out[1]= Integrate[Sin[Sin[x]], {x, 1, 2}]
```

You can, however, ask for an approximate numerical result.

```
In[2]:= N[ % ]
Out[2]= 0.81645
```

Sometimes you just want to do an integral numerically from the start, without having *Mathematica* first try to do it analytically.

```
In[3]:= NIntegrate[Sin[Sin[x]], {x, 1, 2}]
Out[3]= 0.81645
```

`NIntegrate` uses an adaptive scheme, which can cope with weak singularities at the end points.

```
In[4]:= NIntegrate[1/Sqrt[x], {x, 0, 1}]
Out[4]= 2.
```

This finds a numerical approximation to a root of the equation $\cos(x) = x$ near the point $x = 1$.

```
In[5]:= FindRoot[Cos[x] == x, {x, 1}]
Out[5]= {x -> 0.739085}
```

This finds a numerical approximation to the minimum of $x \sin(x)$ close to $x = 2\pi$. The result is a list: the first entry is the minimum value of the function; the second entry is the point at which the minimum occurs.

```
In[6]:= FindMinimum[x Sin[x], {x, 2Pi}]
Out[6]= {-4.81447, {x -> 4.913187501303}}
```

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